TRACEOPS

Autonomous Incident Investigation, Root-Cause Analysis & Safe Recovery

From Operational Signals to Verified Recovery

TEAM PIXELS

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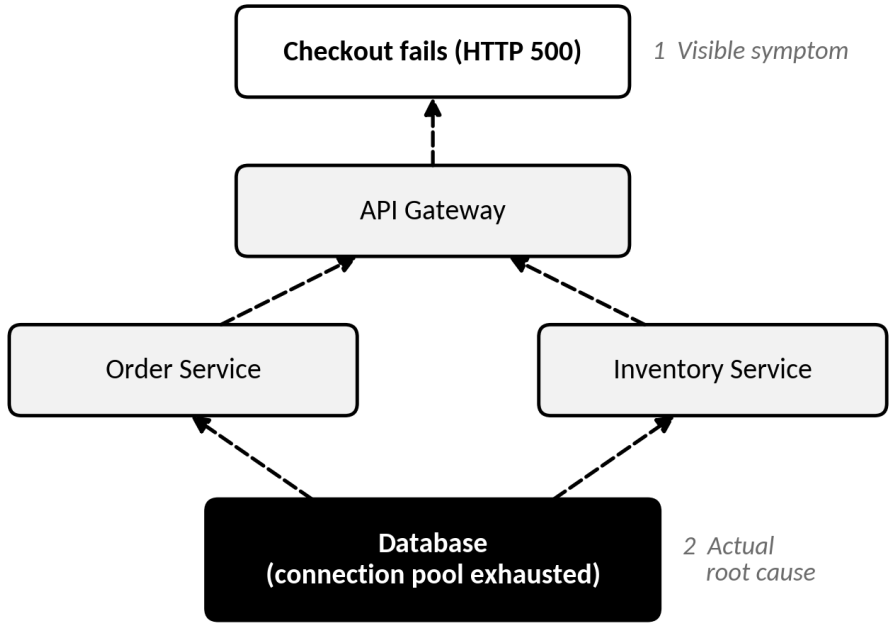
The Problem

Why Is It So Hard to Find the Real Cause of a Failure?

When a system breaks, the error we see is usually just a symptom. The real cause sits somewhere else, and finding it eats up a lot of time.

- Clues are scattered across logs, metrics, traces, alerts, deployments, and tickets.
- One failure sets off others, so the first error is rarely the root cause.
- Connected services make it hard to see the full impact.
- Without enough evidence, teams blame the wrong thing and apply the wrong fix.
- Engineers piece it all together by hand, and acting too early can make things worse.

The core problem: monitoring tells us something is wrong, but not why.
Our goal: find the actual cause, with evidence to back it up.



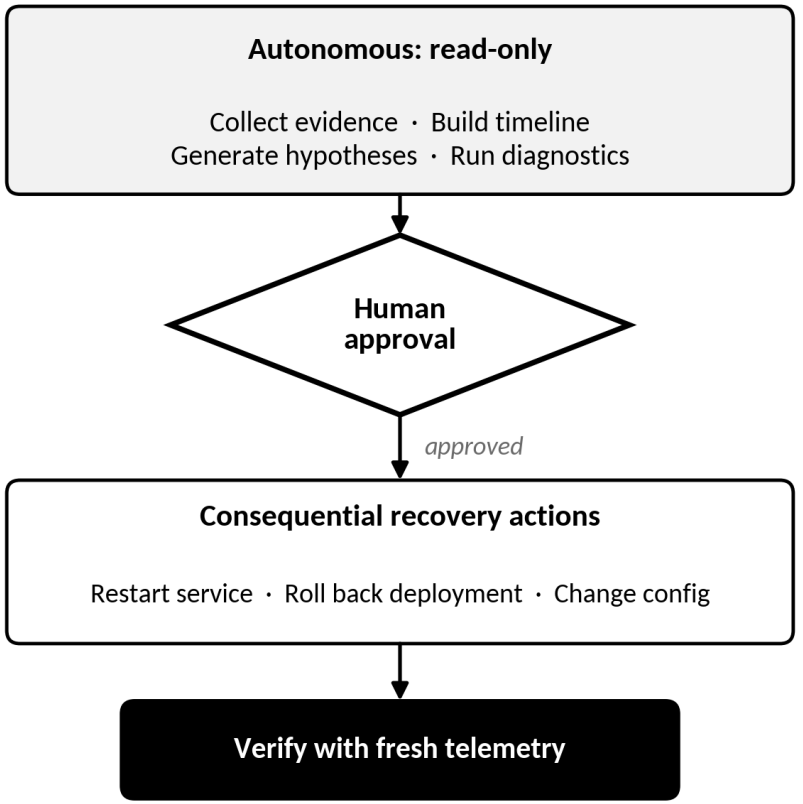
Failure cascades upward, far from where it began

Our Solution

TRACEOPS Investigates Like a Careful Engineer Would

It pulls the evidence together and walks the team through a clear, step-by-step investigation.

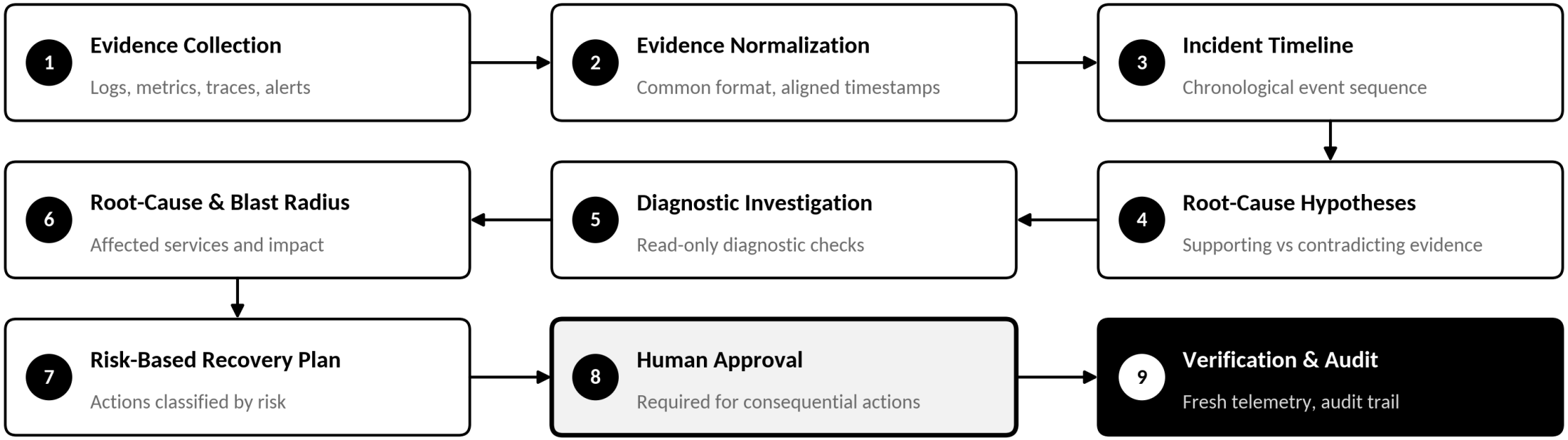
Collect evidence: logs, metrics, traces, alerts, deployments, config changes, and past incidents.
Build a timeline: see what happened before, during, and after the incident.
Test several theories: don't assume the first error is the culprit.
Check the facts: run diagnostics to confirm or rule out each theory.
Learn from history: match the evidence against similar past incidents.
Suggest safe fixes: recommend recovery steps based on the cause and its risk.
Verify the recovery: use fresh metrics to confirm the problem is really gone.



Our key principle: investigation and read-only checks run on their own, but risky recovery actions always need a human's approval.

How It Works

From Incident Detection to Verified Recovery



Technology Stack

Frontend
React, Vite, Tailwind CSS, Recharts

Backend
Python, FastAPI

Database
SQLite

AI Integration
Groq API with Qwen model

Investigation
Deterministic hypothesis scoring and diagnostic tools

Impact & Conclusion

Smarter, Safer, and More Reliable Incident Response

What Teams Gain

- Faster investigations with less guesswork
- A clearer view of the real root cause
- Safer recovery, with approval before risky actions
- Fewer prolonged outages for customers
- Less repetitive manual work for on-call engineers

How We Measure Success

- Root-cause accuracy
- Evidence coverage and traceability
- Investigation time
- False-conclusion rate
- Recovery verification time
- How well it adapts to new evidence

Our vision: TRACEOPS turns incident investigation from a slow, manual hunt into a clear, evidence-driven process. We want to help teams move from reacting to failures towards understanding, preventing, and recovering from them.

INVESTIGATE WITH EVIDENCE. RECOVER WITH CONFIDENCE.

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